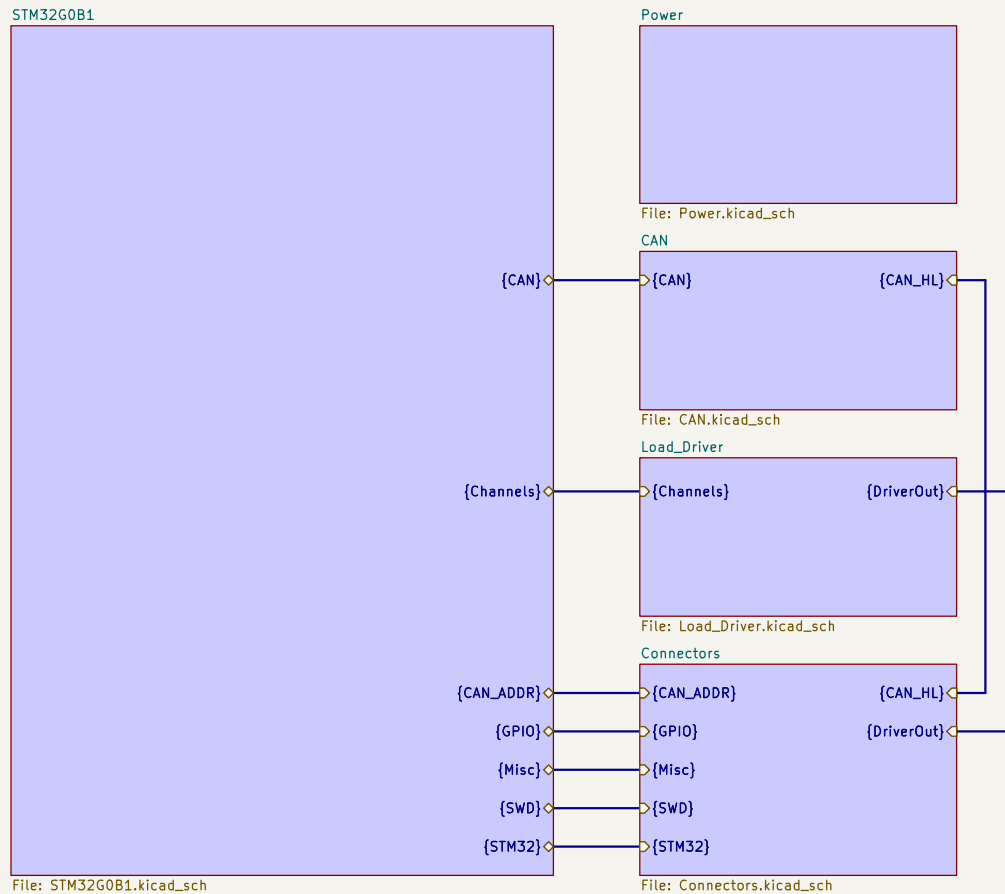


CAN IO Board for NJIT Solar Car

Rev A
06/2026



Author(s): Kostadin Madjovski, Arnav Revankar

NJIT Solar Car

Sheet: /

File: CANIO-Board.kicad_sch

Title: CAN IO Board

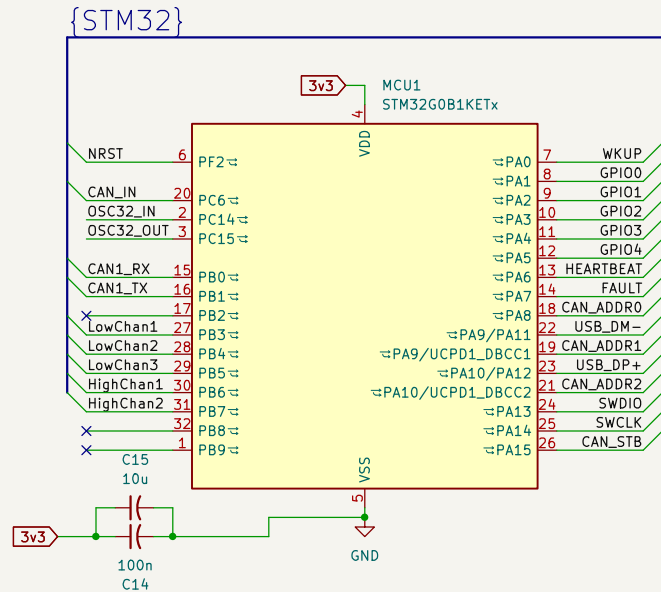
Size: USLetter Date: 2026-06-09

KiCad E.D.A. 9.0.2

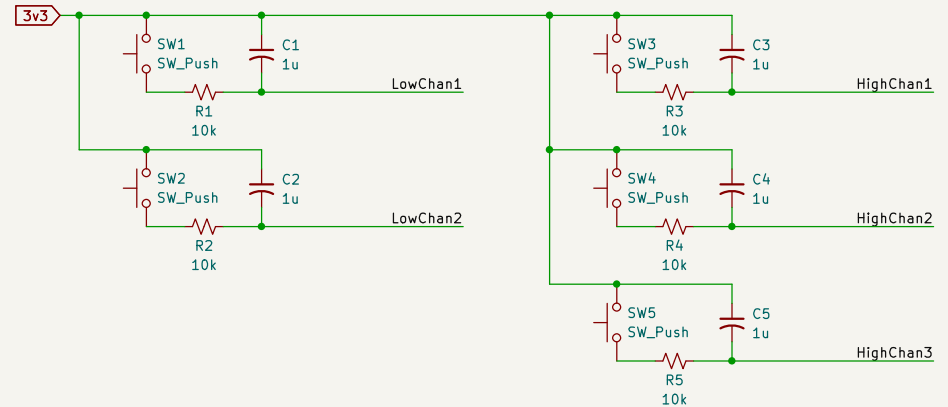
Rev: A

Id: 1/6

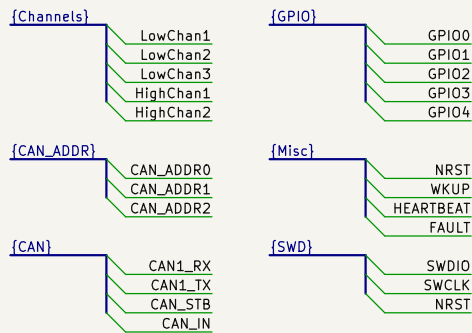
STM32G0B1



Channel Buttons (for debugging)



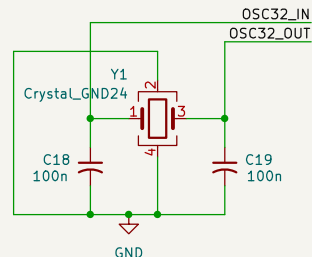
Labels and LEDs



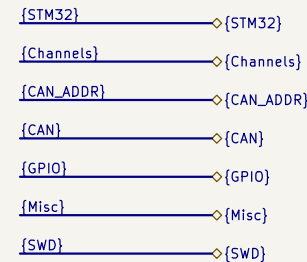
Design Notes

- LowChan and HighChan are gpio outputs, controlling the low and high current load drivers respectively
- All MCU pins are connected via buses, which connect to other hierarchical sheets
- GPIO[0..4] are to be used as needed later. Connector is TBD. The first 3 are capable of GPIO or PWM output.
- Channel buttons (used for debugging) are connected with RC low pass filter for button debouncing. Time constant is 10ms. Look at TI app note for info.
- No external pulldown on the channel inputs because the load driver chip comes with them internally.

32.768kHz Crystal



Global/Hierarchical Labels



Author(s): Arnav Revankar, Kostadin Madjovski

NJIT Solar Car

Sheet: /STM32G0B1/

File: STM32G0B1.kicad_sch

Title: CAN IO Board | STM32G0B1

Size: USLetter Date: 2026-06-09

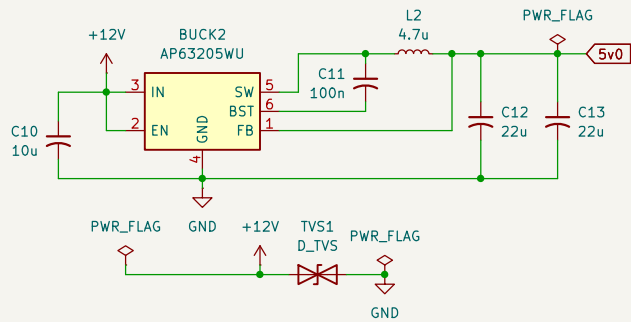
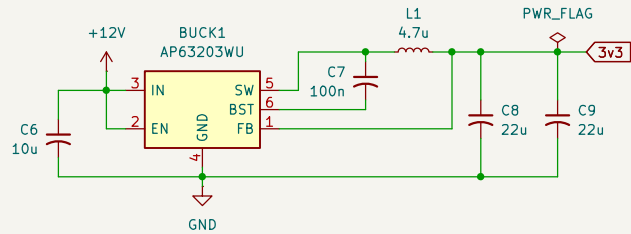
KiCad E.D.A. 9.0.2



Rev: A

Id: 2/6

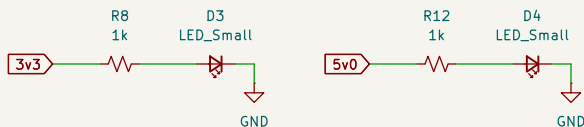
Buck Converters



Design Notes

- All parts of the schematic are from the manufacturer's datasheet.
- TVS diode since the 12V source is exposed
- HOWEVER: What is currently UNCLEAR is the 4.7uH and 2x22uF filtering setup at the output of each IC. Needs to be reviewed.
- Illustration at <https://www.desmos.com/calculator/au2a0t2rlf>

Power LEDs



Author(s): Kostadin Madjovski

NJIT Solar Car

Sheet: /Power/

File: Power.kicad_sch

Title: CAN IO Board | Power

Size: USLetter Date: 2026-06-09

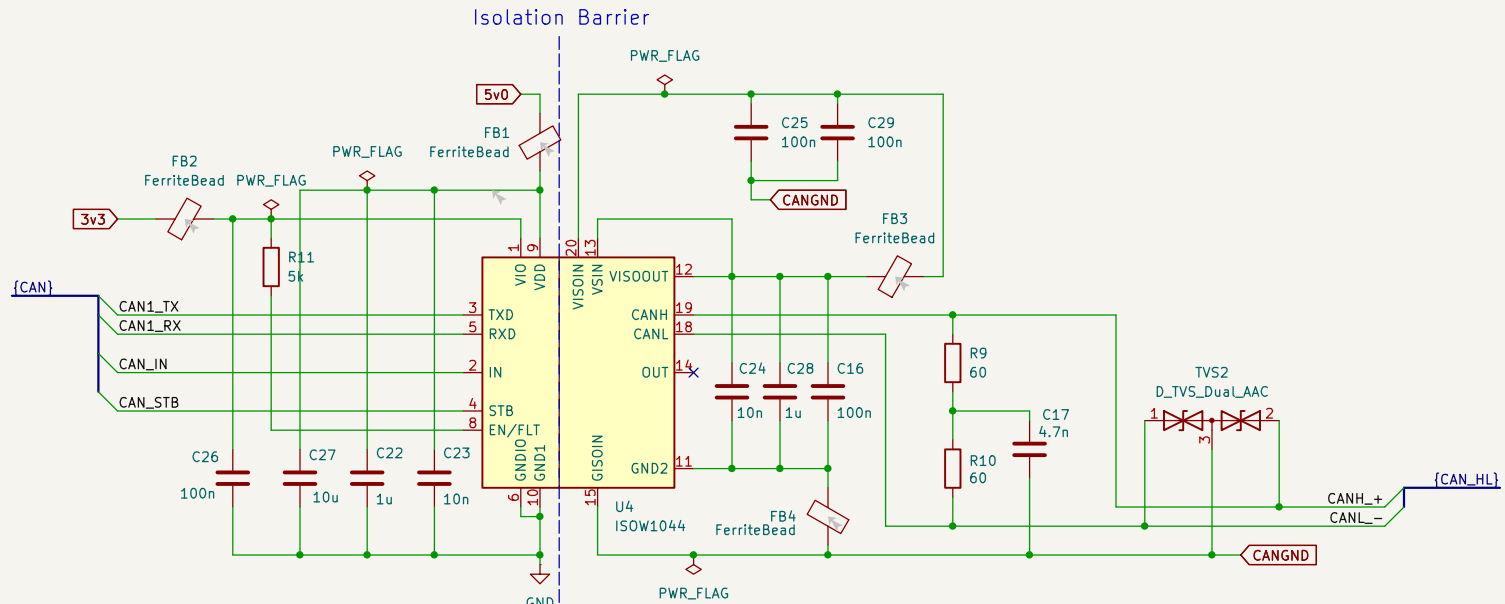
KiCad E.D.A. 9.0.2



Rev: A

Id: 3/6

Isolated CAN Transceiver



Design Notes

- CANH/L+/- are a differential pair (120 ohm differential, 60 ohm odd-mode), and as such trace width, spacing etc will need to be calculated at route-time
- The abundance of ferrites is puzzling, but is what's recommended in the datasheet. The ferrites will all be 0603 package, but the exact bead will have to be determined by the 100MHz impedance, and ZRX curve.
- The ISOW1044 Datasheet recommends the BLM15EX331SN1, which is 330@100MHz.
- Since the CAN lines are exposed, TVS diodes with 12V RWM on both CANH and CANL. For reasoning, check the changelog.
- The 60 + 60R termination is new to me, but is also recommended in the datasheet. This may be swapped for a 120R if need be.
- CAN_IN and STB are both GPIO outputs of the microcontroller. Need to be set manually in code (NOT taken care of by the bxCAN peripheral).

Global/Hierarchical Labels

{CAN} ————> {CAN}

{CAN_HL} ————> {CAN_HL}

Author(s): Kostadin Madjovski

NJIT Solar Car

Sheet: /CAN/

File: CAN.kicad_sch

Title: CAN IO Board | CAN Transceiver

Size: USLetter Date: 2026-06-09

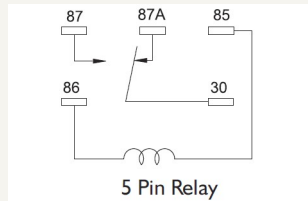
KiCad E.D.A. 9.0.2



Rev: A

Id: 4/6

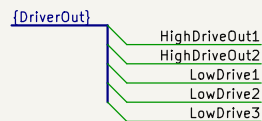
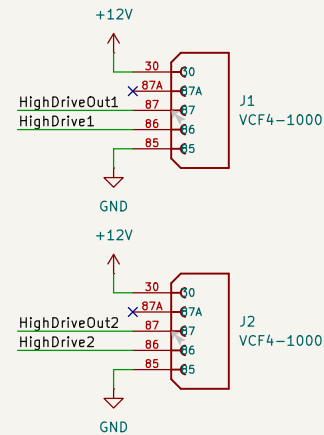
Relay Sockets



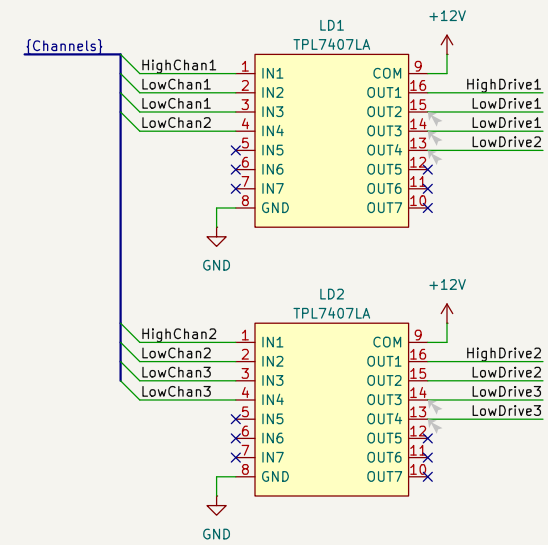
5 Pin Relay

DIN 72552 Standard

- 30: Common
- 85: Coil Ground
- 86: Coil Positive
- 87: Normally Open Output
- 87A: Normally Closed Output



Load Driver ICs



Design Notes

- The load driver ICs are the recommended TPL7407LA's.
- The drivers themselves have only 7 channels, from which we need 2 and have to double up on 3 more.
- Thus, to split the current evenly (put equal load on both ICs), I've set one HighDrive (single channel running a relay coil) per IC, and an additional doubled up LowDrive (1.6A direct draw) per IC. Then, for the last LowDrive, I've split it up between the two ICs.
- This shouldn't cause problems, since both ICs share a common ground, and V_{GS} will be positive for both.

Global/Hierarchical Labels

{Channels} ———— □ {Channels}

{DriverOut} ———— □ {DriverOut}

Author(s): Arnav Revankar
NJIT Solar Car

Sheet: /Load_Driver/
File: Load_Driver.kicad_sch

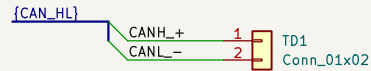
Title: CAN IO Board | Load Driver

Size: USLetter Date: 2026-06-09
KiCad E.D.A. 9.0.2

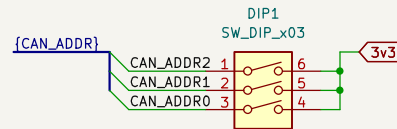


Rev: A
Id: 5/6

CAN Lines (TD connector)



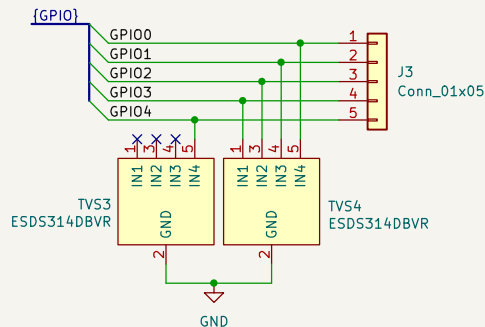
CAN Address Dip Switch



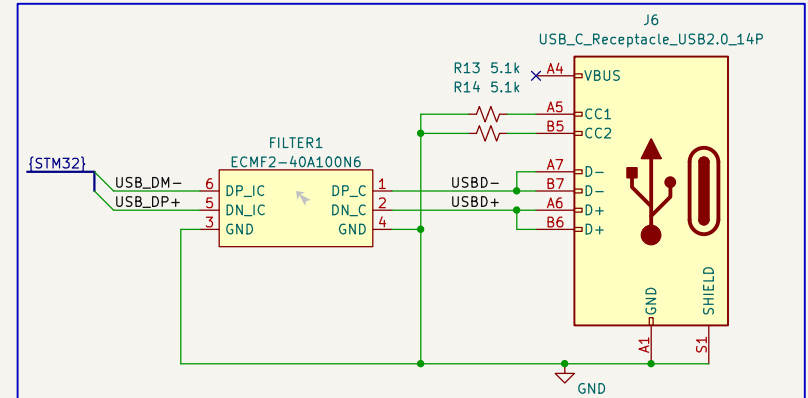
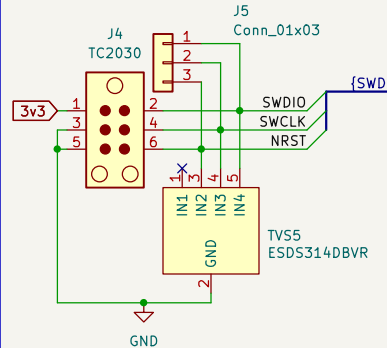
Design Notes

- The CANH and CANL lines are connected to the 2 pin TD connector. As of date the pinout is TBD, might be swapped.
- The CAN Address Dip Switch should control the least significant 3 bits of the board's CAN ID. This will allow upto 8 CAN ID variations, as per the reqs.
- The GPIOs are exposed, and thus have a 4 channel unidirectional TVS diode array. PWR voltage is 3.6V.
- Since the SWD pins are exposed, there's the same 3.6v RWM diodes. But if the stlink programmer operates at 5V (gpio's are 5V tolerant), will need to unsolder this IC.
- Polarity of WAGO connectors is TBD!!! Need to finalize before ordering. (Remove/clarify on this comment when finalized)
- VBUS will NOT power the STM; external 12V is always required. Ground is shared.
- The USB DP DM filter is MODIFIED from the custom library symbol. Restore if modified.
- WKUP is active high with external pulldown. NRST is active low with internal weak pullup.

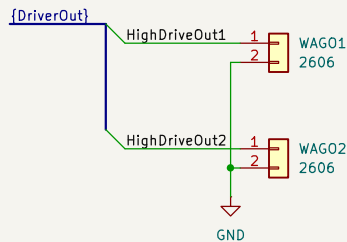
GPIO Connector (TBD)



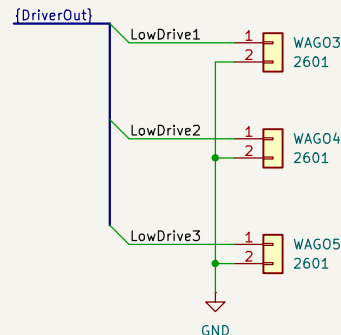
SWD



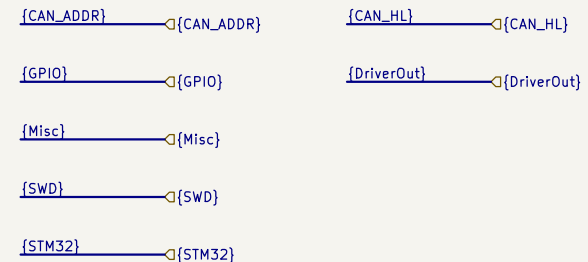
High Current Outputs



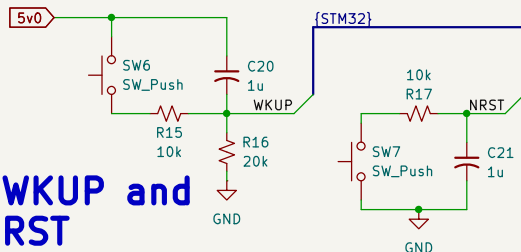
Low Current Outputs



Global/Hierarchical Labels



WKUP and RST



Author(s): Arnav Revankar
NJIT Solar Car

Sheet: /Connectors/
File: Connectors.kicad_sch

Title: CAN IO Board | Connectors

Size: USLetter Date: 2026-06-09
KiCad E.D.A. 9.0.2



Rev: A
Id: 6/6